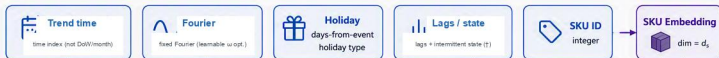


DeepSequence: Hierarchical Attention Forecaster

DS + Softsign + Experts + Context-Aware Mixer

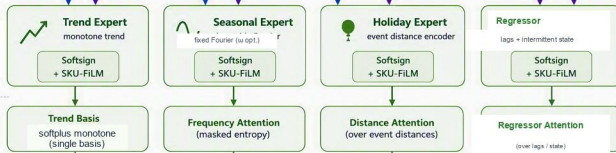
1. INPUTS

Heterogeneous feature groups



2. EXPERTS

Specialized expert networks (monotone: trend, holiday, regressor; seasonal not)



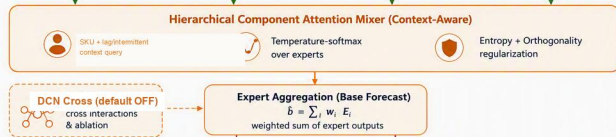
3. INTRA-EXPERT ATTENTION

Capture structure within each expert



4. HIERARCHICAL MIXER

Context-aware mixing across experts



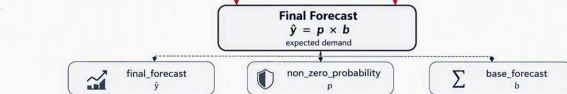
5. FORECAST HEADS

Gated Heads
Decompose forecast into magnitude and occurrence



6. OUTPUTS

Final probabilistic forecast



Legend

Inputs

Experts & Intra-Expert

Mixer & Aggregation

Forecast Heads

Outputs

SKU Conditioning (conditioning path)

Optional / Ablation Path

Notation

E_i output of expert i

w_i attention weight for expert i

b magnitude (base forecast)

p non-zero probability

$\hat{y}$ final forecast

Notes

† Daily: lags 1,2,7

Monthly: lags 1,2,12

(freq. preset / monthly YAMLL)

Quarterly: lags 1,4

Softsign: bounded & stable.

Monotone maps: trend / holiday / regressor only (seasonal not monotone).

Query = SKU :: lag/state; FiLM on expert softsign path.